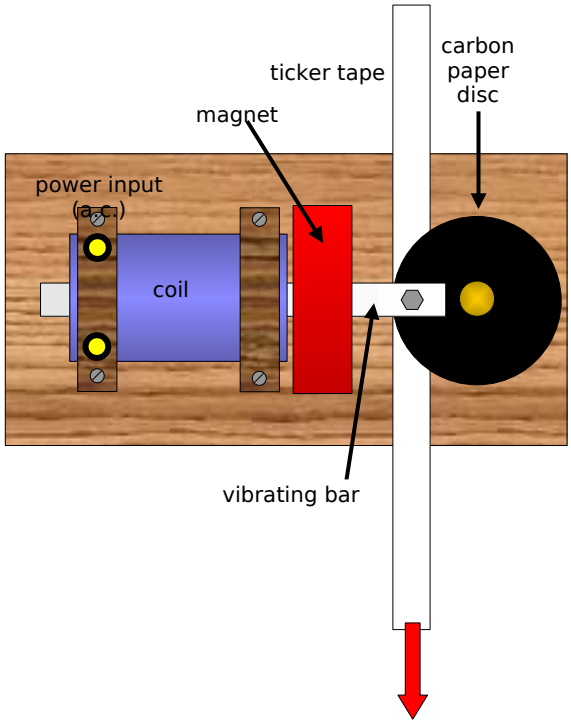
Speed and Acceleration using a Ticker Timer

A ticker timer is a piece of equipment which can be used to measure speed and acceleration.



The ticker timer has a bar which vibrates at a known frequency of 50 times per second. A piece of ticker tape is passed under the bar and a carbon paper disc is threaded between the bar and the ticker tape. There is a small nodule at the end of the bar which produces a dot each time it hits the tape.

The time between each dot is 1/50th of a second, or 0.02 seconds. By measuring the distance between 6 dots (5 spaces) and dividing by the time taken (0.02 x 5 = 0.1 sec) the average speed can be calculated.

Look at the ticker tape below.

Use the following method to calculate the speed the tape was moving.

1. Draw a line through a dot near the left.
2. Draw a second line through the fifth dot after that.
3. Measure the distance between the dots in cm.
4. Calculate the speed in cm per second by dividing by 0.1 seconds (the time it takes to make 5 dots)

Try the two examples below

To calculate acceleration using a ticker timer, you must calculate the average speed of multiple intervals on your ticker tape.

Look at the ticker tape below.

Speed =

Change
in speed =

Calculate the acceleration of this ticker tape by

1. Putting a line through the first dot.
2. Put a line through every fifth dot after that.
3. Use the method on the previous page to find the average speed between the first two lines.
4. Repeat the process for the other 3 intervals to get the average speed for each interval.
5. Find the change in speed between each interval.
6. The acceleration is the change in speed divided by the time between the intervals, in this case 0.1 seconds.

Try to find the acceleration of the ticker tapes below using the method above.